

How effective are trade-related policy instruments (tariffs, subsidies, standards) in internalising environmental externalities?

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Abstract

The tension between the development of international trade and environmental sustainability has become one of the main topics in the international economic policy. Trade related policy instruments such as tariffs, subsidies and regulatory standards are also being used to not only to facilitate economic growth, but also to correct environmental externalities. However, their applicability as far as the internalisation of environmental costs are concerned is debatable particularly with the globalised markets and conflicting national interests. Nevertheless, their usefulness in internalising environmental costs is disputed especially with the globalised markets and divergent national interests.

The paper will focus on the research question: To what extent do trade related policy instruments internalise environmental externalities? The research is a mixed methodology; it involves a theoretical analysis based on the environmental economics and a few case studies which include carbon tariffs, agricultural subsidies and environmental standards in the international trade agreements.

The results indicate that trade policy instruments may be an important part of solving the problem of environmental externalities, but the instruments are very context-specific. Carbon

border adjustment mechanisms and other tariffs demonstrate the potential to reduce carbon leakage, but subsidies usually have ambiguous effects, and in some cases, worsen the situation. Regulatory standards might be good, but they might also be disguised forms of protectionism, which presents problems to the developing nations.

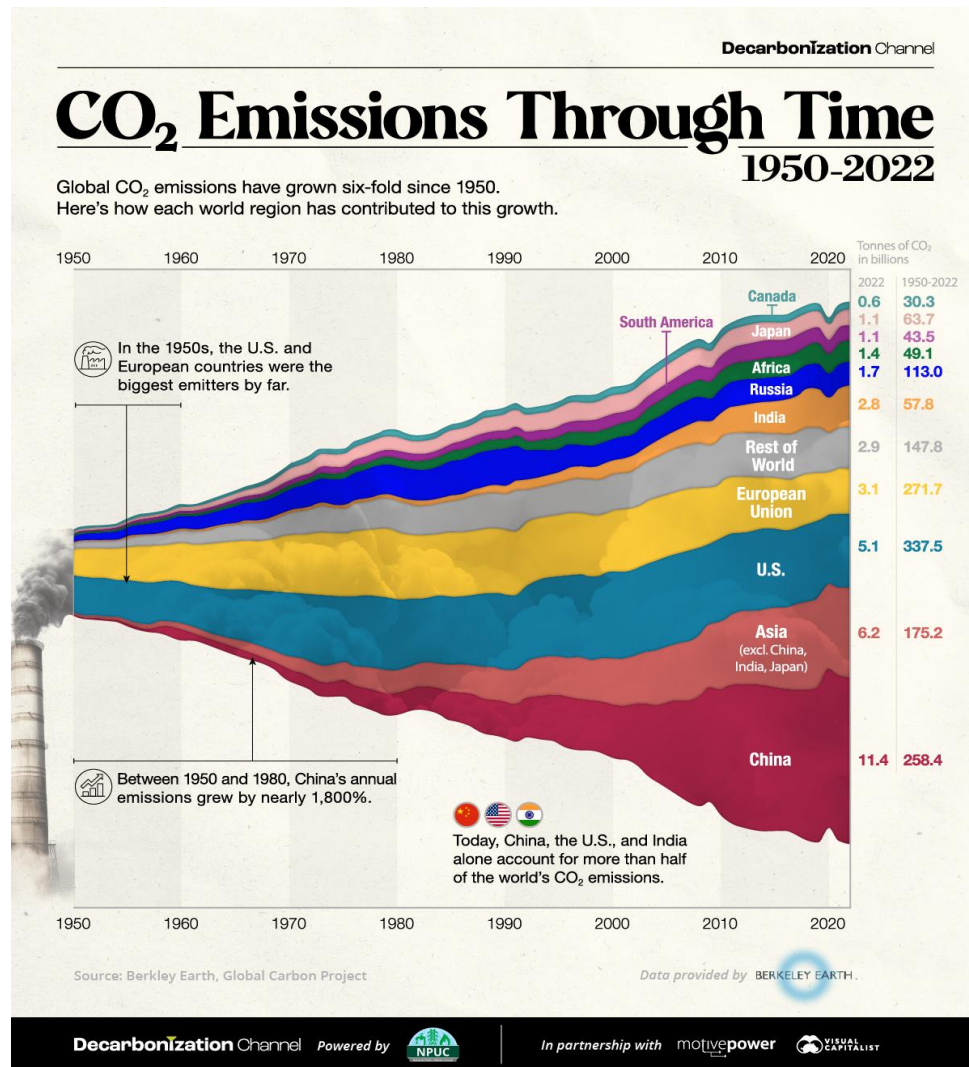
It is concluded in the paper that trade related instruments do not seem to be the right way of completely internalising environmental externalities. To attain the green results, a policy framework that incorporates domestic environmental policies, global collaboration, and open governance is needed.

1. Introduction

The high rate of global trade has in the recent decades greatly reshaped the world economy in that economic growth, technology and better livelihoods have been obtained. The World Trade Organization (WTO) and other institutions have been at the forefront in encouraging trade liberalisation through lowering tariffs, abolition of non-tariff barriers, and making the cross-border flow of goods and services more effective. This has seen international trade to be a significant source of economic integration and development. But, alongside these economic advantages, there has been the rise of increasing worries about the environmental impact of enhanced trade activities.

The increase in global production and consumption trends have led to a broad spectrum of environmental issues; such as climate change, air and water pollution, biodiversity loss and resource depletion. Trade may lead to industrial development which in turn may lead to an increase in greenhouse gas emissions, particularly in those countries where environmental regulations are less strict. In addition, the offshoring of the pollution-intensive industries to the

developing countries, commonly known as the pollution haven hypothesis, begs the question of the environmental justice of worldwide trading. The problems have prompted policymakers and researchers to be much more critical in the study of the relationship between trade and environmental sustainability.



Source : Berkeley Earth, Global Carbon Project

In order to comprehend this relationship better, it is necessary to define the main concepts. Environmental externalities are the unwanted costs or benefits of economic activities,

which are not incorporated in market prices. The trade situation has negative externalities whereby the production or consumption is a source of environmental degradation, but this is not incurred by the producers or consumers but by the whole society e.g. pollution. This leads to market failure whereby the resources are overused or mischanneled without worrying about the environmental damage.

Trade policy instruments on the other hand are instruments which governments can make use of in regulating the movement of international trade. These are tariffs (taxes that are levied on imports or exports), subsidies (financial assistance given to local industries), and standards (regulations that relate to quality, safety or environmental impact of products). Historically, these tools have been employed as a way of safeguarding domestic industries or advancing economic goals. Nevertheless, they have been used to deal with environmental issues more in the recent years. As an illustration, tariffs can be implemented to deter imports which are manufactured in a manner that is harmful to the environment, and subsidies can support green technologies. On the same note, the environmental impact of traded goods can be regulated by environmental standards.

Although it could be theoretically so, the application of trade policy instruments in addressing the environmental externalities is a challenging issue. On the one hand, trade may promote the destruction of the environment by raising production, promoting exploitation of resources, and redistributing pollution in less controlled areas. Conversely, trade may also help to enhance the environment through spreading cleaner technologies and through more effective utilization of resources. The two natures of trade pose a basic policy quandary what to do to make the most out of trade and least out of its adverse effects on the environment.

The key issue, however, is whether such policy instruments as trade-related can effectively incorporate environmental externalities. The process of internalising costs to market prices by making the prices reflect the external costs is known as internalisation to bring the market incentives in line with the social welfare. They may have tools such as tariffs, subsidies, and standards (which sometimes limit their tools by the practicality) but which nevertheless provide potential means of achieving this goal. These are the issues of assessment of environmental harm, enforcement constraints, international trade measures and the fact that the policy can be abused and used to further protectionist goals.

It is in this context that this paper tries to answer the following research question: How effective are tariffs, subsidies and standards in internalising environmental externalities? The question is particularly topical in the current world environment when the environmental sustainability has become the priority of the governments, international organisations as well as civil society. It is imperative to know the strengths and weaknesses of these tools to come up with policies that are capable of meeting the economic and environmental goals.

In this paper, it will be argued that policy mechanisms in dealing with trade can partially internalise the environmental externalities, but this depends on the mechanisms of international coordination and enforcement as well as policy design. The wrong policies can either do nothing or even increase the environmental issues whereas good and coordinated policies can be helpful towards sustainable development. Therefore, effectiveness of these tools is not predetermined but depends on how they are designed and used within more comprehensive frameworks of policies.

The paper will be structured in the following way. The next section gives a theoretical framework, applying the basic concepts of environmental and international economics to explain

how trade policy instruments could be employed to influence environmental outcomes. Then there is the analysis of particular policy instruments: tariffs, subsidies and standards and the analysis of their mechanisms, strengths and weaknesses. The paper will later discuss the pertinent case studies to determine their viability in various situations. Lastly, the conclusion will give an overview of the key findings and offer policy recommendations on how to enhance the application of trade-related instruments in mitigating the environmental externalities.

2. Literature Review

The connection between global trade and the sustainability of the environment has received a lot of discussion in the economic literature. Researchers have discussed the beneficial and adverse environmental effects of trade and the role of policy tools in addressing market failures. This part examines the major theoretical views, the impact of trade-related policies, the impact of major international institutions and agreements and existing gaps in the literature.

(a) Trade and Environment Link.

Trade and the environment are interconnected in a multifaceted way. According to classical theory of trade, free trade will result in efficient distribution of resources due to comparative advantage. Environmental economists maintain, however, that such efficiency does not always factor in the environmental costs, and this might result in possible overexploitation of natural resources.

The Pollution Haven Hypothesis is one of the most eminent theories in this discussion, which states that companies outsource the production that is pollution-intensive to nations with less stringent environmental laws. In a bid to open their economies to foreign capital and

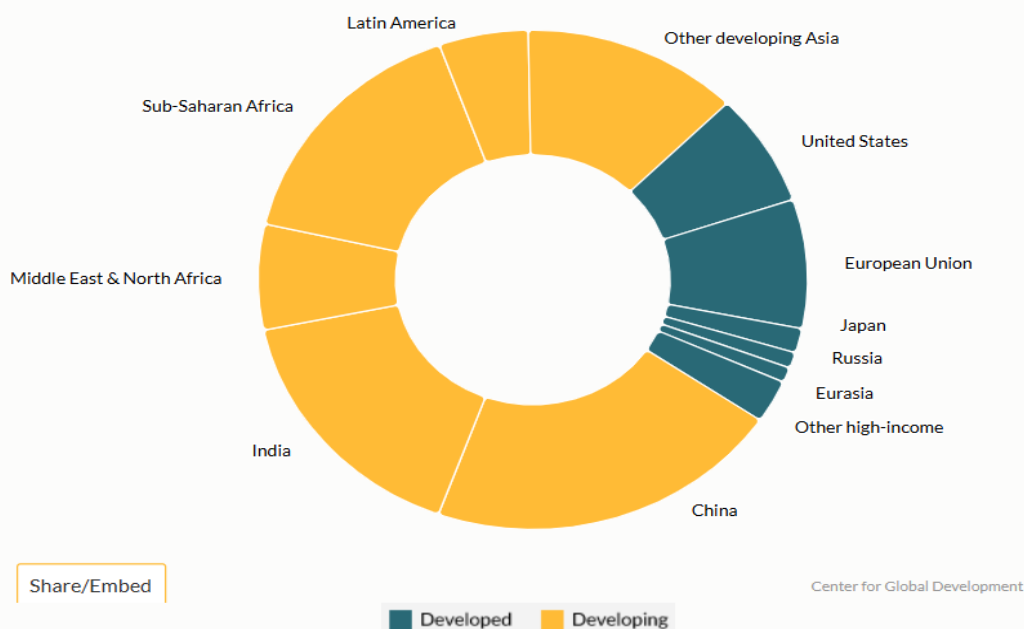
enhance economic development, developing countries can provide lax environmental regulations thus becoming a pollution haven. There is mixed evidence on this hypothesis based on empirical studies. There are studies that pollution intensive industries in fact migrate to less regulated economies but studies have also shown that other factors, like labour costs, infrastructure and access to the market are more important determinants to the investment decision.

Very similar is the so-called Race to the Bottom argument, which holds that nations can intentionally undermine environmental laws in order to be competitive in the global market. Such a vicious cycle may lead to the extensive destruction of the environment, as the governments are more concerned with economic development rather than sustainability. Nevertheless, critics believe that this theory can be excessive in terms of regulation competition. More often than not, nations tend to enact more stringent environmental regulations by either legislative or political pressures within their domestic environment or by international obligations or longer term economic interests.

Another view is the race to the top or California effect in which high environmental requirements in big markets motivate exporters to implement high environmental standards in other nations. This implies that under some conditions, trade has the potential of enhancing environmental results. On the whole, the literature demonstrates that the effects of trade on the environment are not so negative or positive depending on regulatory frameworks, institutional quality, and economic structure.

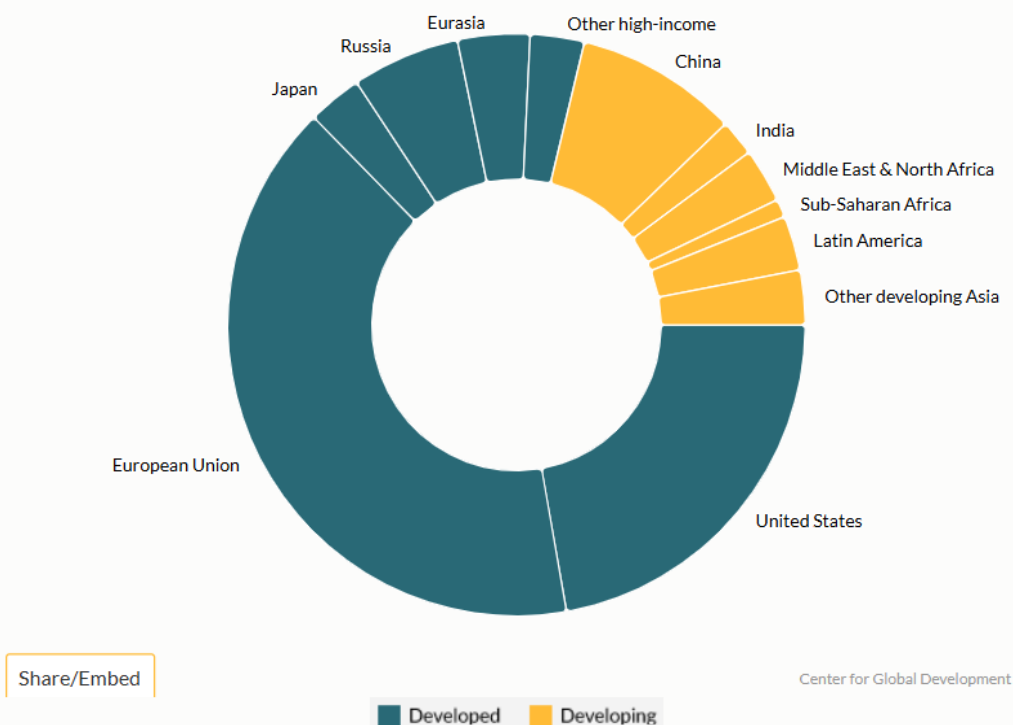
Who's hurt by climate change?

Climate change hits the poor hardest. Poor people living in tropical countries are more exposed to storms and extreme weather, their housing and infrastructure is weaker, and they have less savings or insurance to fall back on when disaster strikes. Source: Social cost of carbon by region, 2015 (RICE-2015).



Who caused climate change historically?

The historical concentration of industry and wealth in developed countries means that they are responsible for 79 percent of the emissions from 1850 to 2011. Source: CO2 emissions excluding LUCF, 1850–2011 (CAIT v2.0).



Source : center for global development

(b) Trade Policies

The policy instruments of trade are usually studied as the means of correcting the market failures related to environmental externalities. The theory of environmental economics claims that the negative externalities (e.g., pollution) lead to socially inefficient results since neither the producer nor the consumer bears the complete cost in their behavior. It is thus necessary to internalise these external costs by policy interventions.

Some of the significant instruments in this connection are tariffs, subsidies, and regulatory standards. Environmental tariffs like carbon border adjustment systems are geared towards leveling the playing field through the imposition of costs on imported goods that are produced under less strict environmental standards. In principle, these actions should decrease carbon leakage and encourage cleaner production practices in the world. Their efficacy is however disputed because of difficulties in measuring embedded emissions correctly and meeting the international trade regulations.

Another useful tool is the subsidies, especially in encouraging environmental friendly technologies and industries. As an example, renewable power subsidies can hasten the process of moving to low-carbon economies. However, the literature identifies that there are also unwanted consequences of subsidies. The environmental degradation can be increased by environment-damaging subsidies, e.g., subsidies of fossil fuels or intensive farming. Moreover, subsidies can pervert market competition and cause inefficiencies when not targeted appropriately.

Environmental and technical standards are regulatory standards that are popular in regulating the environmental impact of traded goods. These standards can be powerful in providing the minimum environmental quality and safety. They are however, frequently condemned as a non-tariff barrier to trade especially when they disproportionately raise the costs

of compliance on the developing countries. The literature underlines the importance of the design and implementation of these policies in ensuring their effectiveness.

To conclude, despite the fact that trade policy tools may be useful in addressing the issue of environmental externalities, their usefulness depends on several factors, such as transparency, enforcement capacity, and compatibility with overall environmental policies.

(c) Key Institutions & Agreements

The international institutions and agreements are very important in determining the interplay between trade and the environmental policy. World Trade Organization (WTO) offers the most predominant legal framework of international trade. Its regulations are made to encourage non-discrimination, and minimize trade barriers, yet also provide an exception of some environmental exceptions. An example of this is that countries can place trade restrictions on the ground that they are protecting human, animal, or plant life, but the restriction cannot be in a discriminatory or protectionist form.

Nevertheless, there is usually a conflict between the trade liberalisation and environmental protection in the WTO set-up. The critics claim that the WTO rules can limit the application of trade measures to achieve environmental interests, especially when the measures are seen to be veiled protectionism. Cases of dispute settlement have demonstrated the challenge posed by reconciling trade requirements with the environment.

Besides trade institutions, there are increased global environmental agreements. The Paris Agreement is a milestone in international action against climate change by curbing the rise of global temperatures. Despite the fact that the agreement is mainly concerned with the national commitments to cut greenhouse gas emissions, it has great implications on the trade policy.

Examples include implementing trade-related policies by countries to meet their climate goals, including carbon pricing or environmental standards.

The relationship between trade and environmental agreements is a developing research topic. There are those scholars who support the idea of more integration of the two regimes, citing that co-ordinating policies can only make them more effective and lessen the conflicts. Some warn that overlapping jurisdiction and different priorities can lead to inconsistencies in policies.

(d) Gaps in Literature

Although a great deal of research has been carried out, there are still some gaps in the literature on trade-related policy instruments and environmental externalities. The absence of a unanimous opinion on the effectiveness of these instruments, in general, is one of the most important problems. Whereas other studies emphasize their potential to internalise environmental costs, other studies indicate their minor effects and unintended impacts.

The major problem is that it is hard to quantify externalities to the environment and assign change to certain policy actions. The change in technology, economic growth and domestic regulations are only some of the many factors that affect environmental outcomes. It becomes difficult to separate the impacts of trade policies and to make conclusive findings.

The other gap of importance is the variation in the effects of trade policies on the developed and developing nations. Institutional capacity, mechanisms of enforcement and financial resources available to developed countries generally enable them to enforce environmental policies effectively. Conversely, developing nations can be constrained by certain factors such as absence of administrative capability, higher compliance expenses and other

conflicting needs in development. As a result, the related environmental policies may not be homogenous which can lead to global inequalities.

Furthermore, there is a paucity of studies carried out on how effective these instruments are in the long run, to generate sustainable results. Much of the research is on the short-run effects, and does not examine larger structural changes in the economy or the environment. More interdisciplinary studies which combine the perspectives of economics, political science, and environmental studies are also needed.

Finally, the literature highlights the risk of using the policies in a wrong manner, whereby the protectionism is masked into environmental protection. This is one of the issues that raise concerns on fairness and legitimacy in the international trading system to the developing countries which may be overworked by such actions.

Conclusion of Literature Review

Overall, the literature has offered significant pieces of evidence concerning the complexity of the issues of trade and environmental sustainability. Trade may be damaging and beneficial to the environment, but policy interventions are important in influencing the results. The means of internalisation of environmental externalities, whose effectiveness is not yet fully understood and context-specific, can be provided by the tools of trade (tariffs, subsidies, and standards). This dynamic is further affected by international organizations such as WTO and agreements such as the Paris Agreement in terms of which we must apply coordinated and balanced strategies. The literature gaps that exist indicate the need to conduct further studies especially in the differentiation of the effects across nations and the long-term efficacy of policy actions.

3. Theoretical Framework

This part gives the theoretical basis on which the effectiveness of trade-related policy instruments to internalise environmental externalities can be analysed. It is based on some major concepts of the environmental economics and the international trade theory, and it revolves around the market failure, internalisation mechanisms and the trade policies as corrective mechanisms.

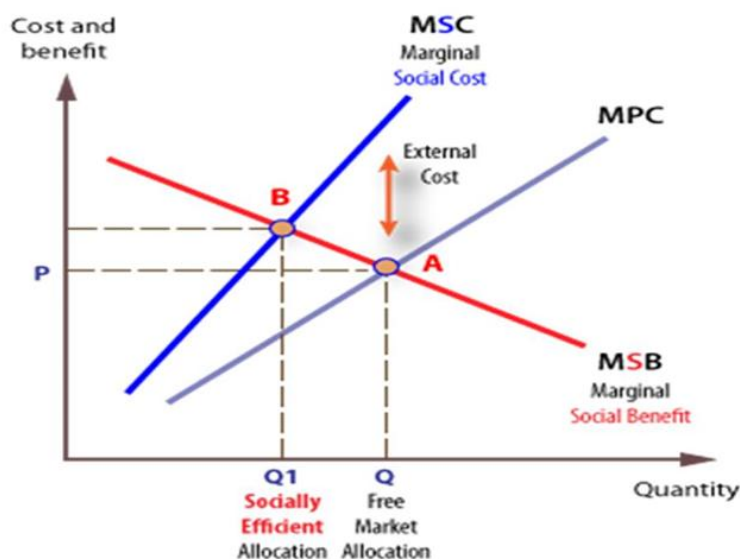
(a) Environmental Externalities and Market Failure

Environmental externalities are the costs or benefits which economic activities may inflict on third parties not reflected in market prices. These externalities are in most instances that are negative (e.g. pollution produced in production or transportation) as it applies to this study. An example is a manufacturing company dumping greenhouse gases or disposing waste in water bodies without incurring the entire social cost of such actions. Consequently, the social costs and the private costs part ways, causing ineffective allocation of resources.

This is a typical market failure where free market does not achieve socially optimal results. In a perfectly competitive market, prices should be able to indicate all the costs and benefits involved and the resources should be allocated efficiently. But in the presence of environmental externalities, the marginal private cost of production (MPC) is less than the marginal social cost (MSC). Therefore, companies will generate a volume of output that is above the socially optimal volume and prices will be artificially low.

$$MSC=MPC+MEC$$

In this case, MEC is the marginal external cost to the society. Socially optimum level of production is where the marginal social cost is equal to marginal social benefit (MSB), however, when externalities exist, markets would over produce and lead to environmental degradation.



Source : www.economicsonline.co.uk

This issue can be accentuated in the framework of global trade. Liberalisation of trade promotes production and consumption in industries that may be resource intensive or intensive in pollution. Specialisation in such countries can lead to increased harm to the environment especially when environmental standards are ineffective. Moreover, variations in the standards of the environment in different nations may corrupt the pattern of trade, whereby a company would want to move their production to where the cost of regulations is lower.

Therefore, the environmental externalities do not only result in domestic inefficiencies but also cause spill-over effects across borders, hence this is a global issue. To deal with these externalities, policy interventions are needed that can bring the private incentives and social welfare on par.

(b) Internalisation Mechanisms: Pigovian Taxation Logic.

Among the most popular methods to rectify the externalities of the environment is derived on the principle of Pigouvian taxation which was formulated by economist Arthur Pigou. The overall point is that the marginal cost of pollution is a tax that the government can establish, which would internalise the externality and would restore market efficiency.

$$t=MEC$$

The marginal external cost can be taxed (t) to make the social cost equal to the marginal cost of production. This moves the supply curve upwards decreasing output to the socially optimal and raises prices to cover the environmental harm. Consequently, companies are motivated to use cleaner technologies or cut down on their production in dirty industries.

Pigouvian taxes are very efficient and flexible. They do not force firms to choose the optimal method of reducing the emissions, as opposed to direct regulation, and promote cost-efficient solutions and innovation. A well-known example of such practice is carbon taxes, which are extensively debated in the framework of combating climate change.

Nevertheless, there are various difficulties in applying Pigovian taxes in a real world. Proper estimation of the marginal external cost of environmental degradation is not that easy especially in complicated problems like climate change. Moreover, political and economic factors can cause constraints on the desire of governments to introduce such taxes, in particular, when it concerns the competitiveness in the international market.

The other weakness is that of international trade. When stringent environmental taxes are paid in one country and other countries do not pay them, then local companies will incur increased costs of production thus making them less competitive. This may cause leakage of carbon to other countries having less stringent regulations, which negate the policy.

To solve these issues, economists have considered other or complementary systems, such as tradable permits, regulation standards, and policy instruments of trade. Such methods are meant to accomplish the same by ensuring that environmental costs are included in economic decision-making.

(c) Trade Policy Link: Tariffs, Subsidies, and Standards as Remedial Means.

The policy instruments that are related to trade could be conceived as expansion or adaptation of internalisation mechanisms at the international level. Although traditional environmental policies are applied within the national scope, the trade policies can affect the cross-border behaviour and, thus, this aspect is especially important to global environmental externalities.

Tariffs may serve as corrective measures when they are set in a manner that they capture the environmental cost of imported goods. As a case in point, a nation can put in place a carbon tariff on imports manufactured in jurisdictions that have lax environmental rules. This in turn puts the playing field at par with imported products since they are subjected to environmental costs equal to domestic products. Such tariffs theoretically can decrease carbon leakage and encourage foreign manufacturers to be cleaner.

On a theoretical level, environmental tariffs may be considered quite similar to Pigovian border taxes. They would internalise external costs related to production processes that are not within the jurisdiction of the regulating country. Nonetheless, their effectiveness is subject to proper measurement of environmental effects, adherence to the international trade regulations, and avoidance of protectionist abuse.

The other significant policy instrument is subsidies, especially when it comes to achieving positive environmental results. Green subsidies, including renewable energy, energy efficiency or sustainable agriculture, can be used to correct market failures related to the underinvestment in technologies that are environmentally friendly. Subsidies can be used to reduce the cost of clean alternatives thus making them more acceptable and contribute to the shift to more sustainable production systems.

Theoretically, negative as well as positive externalities can be tackled through the use of subsidies. Whereas Pigouvian taxes concentrate on penalising activities that are harmful to the environment, subsidies are provided as a reward to favourable behaviour to the environment. Their success, however, depends on their appropriate targeting and design. The bad design of subsidies can result in inefficiencies, fiscal costs, or unintentional environmental damage. An example of this is that fossil fuel or intensive agriculture subsidies may intensify environmental degradation, and subsidies must be adjusted to meet sustainability goals.

The most direct way of controlling the externalities of the environment may be through standards and regulations. These are the environmental standards, the technical regulations, and certification requirements which establish minimum standards of environmental performance. When applied to trade, these standards may affect local and international manufacturers by limiting the access of goods that fail to comply with outlined standards to the market.

In theory, standards may have the same effects as Pigouvian taxes, by restricting the conduct of harm to the environment and ensuring adherence to the environmental goals. They, however, do not give companies flexibility to adopt the most cost effective mode of compliance as with market based instruments. Also, standards can act as obstacles to trade especially to developing nations that might not be in a position to attend to high quality standards.

Regardless of such constraints, the standards may be instrumental in dealing with information asymmetries and maintaining quality and safety of products. As an illustration, eco-labelling initiatives can give the consumer details on the environmental influence of products and therefore make informed decisions. This is able to provide market incentives to producers to be sustainable.

Integration of Concepts

Theoretical framework recognises the fact that trade policy instruments, such as tariffs, subsidies and standards will be useful as a tool of internalising environmental externalities in a globalised economy. All the instruments have their pros and cons and can be utilized with the assistance of the quality of its design and execution.

Tariffs are particularly easy to use in solving cross-border externalities and preventing regulatory arbitrage, but they have to be carefully tuned to prevent trade conflict. The use of subsidies can hasten the use of green technologies but needs to be focused to prevent inefficiency. Standards give certain regulatory guidelines but can be expensive to comply with and can limit market access.

It is worth noting that these tools do not seem to have any exclusivity in their relation to each other and are generally most effective when used together. The following are some examples of how a comprehensive policy can be achieved: Carbon pricing, border adjustments, green subsidies and environmental standards which can be supported by international collaboration and institutional structures.

Conclusion of Theoretical Framework

Lastly, the theoretical framework demonstrates that the externalities of the environment is a critical market failure that needs a policy intervention. The principle of Pigouvian taxation

provides a logistic background of internalisation of these externalities but its application in global setting requires customisation with the trade-related policy tools. The tariffs, subsidies, and standards offer viable means of integrating the environmental costs in the market decision making, but they have been shown to do so differently in the effectiveness of the design, enforcement and international coordination. It is this framework that gives us the foundation of the evaluation of these instruments in practice as seen in the following sections of the paper.

4. Policy Instruments Analysis.

(A) Tariffs

Trade policies have historically employed tariffs as a means to safeguard local industries and raise revenue that can be used by the government. They have however come in the recent years to be viewed as instruments to deal with environmental externalities especially with regard to climate change. The purpose of environmental tariffs and carbon tariffs in particular is to internalise the environmental cost of production in the price of traded goods, assisting in the internalisation of negative externalities.

The Carbon Border Adjustment Mechanism (CBAM) proposed by the European Union can be considered one of the brightest illustrations of such actions. The CBAM is meant to levy a price on imports of some commodities (steel, cement, aluminium, etc.) as per the emissions produced during their manufacture. The aim of this mechanism is to harmonize the carbon price of goods imported with that of the goods produced in the EU, where domestic producers already face carbon prices as a result of emissions trading systems.

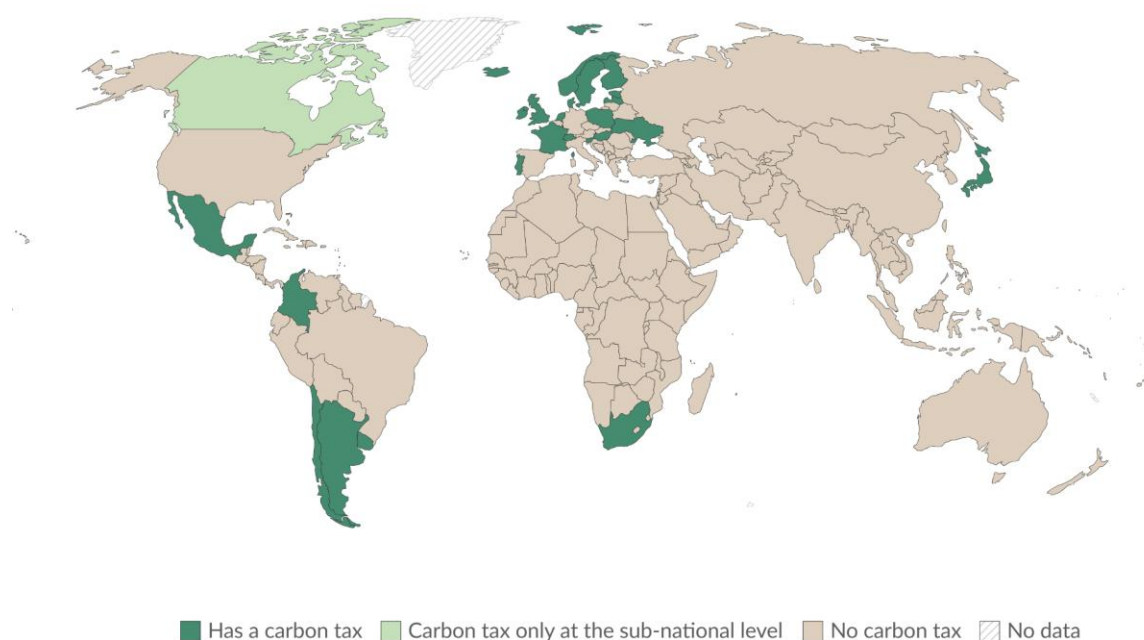
Theoretically, environmental tariffs can be viewed as a continuation of the Pigovian border tariff. When nations implement domestic environmental policy, e.g. carbon tax, producers

will incur an increased cost, which could lower their competitiveness in international markets. This may cause carbon leakage, with production moving to nations with less stringent environmental legislation, unless there are equivalent actions on imports. Governments seek to eliminate such leakage by the application of tariffs, which indicate the amount of carbon content in imported products and by doing so, the governments seek to ensure that the costs associated with the environment are internalised irrespective of where production is carried out.

Which countries have a carbon tax? 2024



A country is marked as having a carbon emissions tax instrument if at least one sector has implemented one.



Data source: Dolphin and Merkle (2024)

OurWorldinData.org/co2-and-greenhouse-gas-emissions | CC BY

Source : Dolphin and Merkle (2024)

Advantages of Environmental Tariffs

The possible reduction of carbon leakage is one of the main benefits of environmental tariffs. Without such measures, companies could move their production to jurisdictions with less

stringent environmental regulations, which will hinder the global fight on reducing emissions. The tariffs keep the domestic environmental policies effective by ensuring that the price of carbon is equal between domestic and foreign producers.

Moreover, cleaner production practices can be promoted in the world through environmental tariffs. Buyers of environmental tariffs on markets that require exporting companies can be encouraged to use green technologies and minimise their carbon footprint. That way, tariffs may have a spill over effect, facilitating the environmental improvement across national borders. This is especially needed in dealing with international environmental issues like global warming, where international cooperation is needed.

The other benefit is that tariffs may earn the government revenue, which could be re-invested in green activities, e.g. development of renewable energy or adaptations to climate changes. This is a two-fold advantage: not only is market failure corrected but also helps in the shift to a sustainable economy.

Moreover, the tariffs on the environment can also help in creating a level playing field among the industries within a country. Companies whose activities are controlled by stringent environmental laws tend to be expensive to produce. In the absence of border adjustments, they can be at a disadvantage compared with international rivals. Tariffs also serve to correct such an imbalance, and the risk of unfair competition, based on varying environmental standards, is minimized.

Limitations and Challenges

Although environmental tariffs have the potential advantages, they have quite a number of limitations. Among the most significant issues is associated with their compliance with the regulations of the World Trade Organization (WTO). The principles of WTO focus on the non-

discrimination principle and the need to avoid protectionist practices. Environmental tariffs can also be questioned as it is seen as discriminatory or as veiled barriers to international trade. To ensure such tariffs meet the WTO requirements, they must be carefully designed, transparent and consistently applied.

The other significant constraint is the imbalanced effect on the third world countries. Most developing economies depend on exporting carbon-intensive products and might have no technological and financial resources to switch to the cleaner production process. Environmental tariffs that are imposed by the developed countries can consequently lower their competitiveness in terms of exports thus potentially limiting economic growth and furthering global inequalities. The critics claim that these steps will transfer the burden of environmental adjustment to those countries that have had the least contribution to the past emissions.

Besides, carbon content of imported commodities is a technical difficulty to measure. The methods of production processes demand a lot of information and standardised procedures to accurately determine the emissions that the process entails, which are not necessarily easily accessible. This process is further complicated by the differences in production methods, sources of energy, and supply chains. Measurement errors may render tariffs ineffective and may cause conflict among trading partners.

The other risk is that the environmental tariffs can be used as a kind of protectionism. Governments can also impose tariffs on environmental reasons, but their main objective is to shield domestic industries against international competition. This abuse may destroy confidence in the international trading system and prompt retaliatory action which may culminate into trading wars.

Also, environmental tariffs are only a part of the large problem. Although they are able to affect trade flows and production, they do not dictate domestic consumption or production in countries that export. As a result, their net contribution to the emissions at the global scale can be minimal unless it is supplemented by the other policy instruments, such as the pricing of carbon domestically, global agreements and technological collaboration.

Critical Evaluation

There are a number of factors that determine the success of environmental tariffs in achieving externalities of the environment to the internalisation. Firstly, it is essential that the policy should be designed. The tariffs should be founded on the correct calculation of the environmental costs and be realized in a non-discriminatory and transparent way. Second, there should be coordination of various countries to minimize the conflicts and establish uniformity among the countries. Unilateral action can be of less assistance and can more often lead to conflicts.

Third, the tariffs can be augmented to be more effective when they are combined with other policy frameworks. An example of this is that carbon tariffs which are supplemented with domestic emissions trading systems, environmental laws and green subsidies can be designed to create a more holistic sustainability strategy. Fourth, the developing countries should be assisted to address the issues of equity. This can be either in the form of financial assistance, technology transfer and capacity building to allow these countries to adapt to the stricter environmental factors.

In practice, the environmental tariff such as the CBAM in the European Union is an important step towards integrating the environment into the trade policy. Nevertheless, they do

not exist as an independent solution. The effectiveness of their implementation will depend on the extent of their implementation and the existence of collaborative international efforts.

Conclusion

To conclude, environmental tariffs are a great but imperfect way of internalising environmental externalities in a globalised economy. They will be in a position to reduce carbon leakage, promote cleaner production and support domestic industries that are highly regulated by environmental regulations. Their efficiency, however, is limited with issues of trade regulations, measurement problems and equity issues. Although tariffs may be very important, they would need to be well-planned and supplemented with the other policy tools to ensure that there are meaningful and sustainable environmental results.

(B) Subsidies

A key tool of environmental and trade policy is subsidies, which are used by governments to shape the production and consumption behaviour by reducing the cost or offering financial incentives. Green subsidies are created as a measure of environmental sustainability to stimulate the use of renewable energy, energy-efficient technologies and processes of production that are friendly to the environment. Contrary to tariffs, which punish environmentally harmful actions, subsidies are used to promote positive externalities by promoting those activities that bring about social and environmental positive results.

Theoretically subsidies can be explained as a solution to market failure related to under-investment in clean technologies. Renewable energy sources like solar and wind energy are more likely to be innovated with high initial costs, uncertain returns, and knowledge spillovers that provide greater benefits to society as a whole than to individual companies. This means that the

level of investment that is socially optimal might not be attained by the private firms. Governments can help fill this gap by offering financial assistance and hastening the process to a low-carbon economy.

The recent example is the Inflation Reduction Act (IRA) passed by the United States in 2022. The IRA incorporates significant subsidies to produce renewable energy, electric vehicles, battery production, and clean technology. It provides tax credits, grants, and incentives to cut down on greenhouse gas emissions and enhance the capacity of domestic industries. The policy will be seen as a strategic application of subsidies not just on environmental purposes but on economic competitiveness of new green industries.

The benefits of Green Subsidies.

Green subsidies have the ability to encourage the use of clean technologies; this is one of the greatest benefits of green subsidies. Subsidies can make renewable energy and sustainable practices more competitive by lowering the price of these alternatives compared to the traditional system, which depends on fossil fuels. This can hasten the transition towards the low-carbon energy, which will help reduce the emissions and make the environment sustainable.

Another important role is played by subsidies as a way of promoting innovation. Clean energy development in order to develop a technology in the industry in most cases takes a large amount of research and development (R&D). Government assistance will decrease financial risk faced by firms, which will allow them to invest in new technologies and scale up the production. This may cause the reduction of the costs over the time as the effect of economies of scale and learning-by-doing. To illustrate, the recent reduction in prices of solar panels all over the world has been partially blamed on long-term government subsidies in major markets.

The other benefit is that the subsidies may assist in the establishment of new industries and jobs. The shift towards a green economy would necessitate the creation of new industries, i.e., renewable energy, electric mobility, and energy storage. Governments may promote economic growth by supporting the industries and at the same time solve environmental problems. Green subsidies, in this sense, are useful to both the environmental and industrial policy.

Moreover, subsidies may be used to deal with positive externality of environmental goods. To give examples, the extensive use of clean technologies can help to decrease air pollution, the level of health among people, as well as decrease the cost in the long-term environment. Subsidies assist in aligning the private incentives with social welfare in that these benefits are not entirely taken in by the private firms.

Moreover, subsidies may be more acceptable politically as opposed to taxes. Environmental taxes raise costs incurred by consumers and producers but subsidies offer financial relief and are easy to apply in most situations. This is especially necessary in terms of securing popular and political backing of the environmental policies.

Limitations and Challenges

Green subsidies have various limitations though they are advantageous in many ways. Market distortion is one of the major issues. Subsidies disrupt market price, which may result in resource depletion. Without proper targeting of subsidies, there is a risk that the subsidies could favor industries or technologies that will not be viable in the long term and this will lead to wastage in public resources.

The risk of subsidy wars is another big problem. When nations give extensive subsidies to their domestic companies, other nations may retaliate with the same to be at par with the

world. This may give rise to a spiral of increasing government subsidies, distortion of international trade and cause strains between trading partners. Such subsidies as those in the US Inflation Reduction Act, e.g. have given other economies such as the European Union apprehensions that such policies will drive investment out of its own industries.

Governments can also be burdened with a massive fiscal cost because of subsidies. Massive subsidies demand a lot of funding by the government which can be a burden on the government budget especially in developing nations. This casts doubt on the sustainability of these kinds of policies particularly when there are no long term payoffs.

In addition to this, it can be poorly designed or politically influenced resulting in poor policy allocation and inefficiency. Governments can also use subsidies to distribute the subsidies based on political grounds, instead of economic or environmental performance. This may lead to subsidizing less efficient technologies or companies, minimizing the effectiveness of the policy.

The other weakness is that subsidies do not necessarily result in intended environmental outcomes. As an illustration, subsidies on some biofuels have been blamed as causing deforestation and rising food prices. Likewise, without regulations or other pricing schemes that will deter such polluting activities, subsidies may not be effective in curbing emissions.

In the global sense, subsidies may cause trade conflicts. Some forms of subsidies are considered to be trade-distorting and can be challenged under the rules of the World Trade Organization (WTO). This puts legal and institutional limits on the utilization of subsidies, especially in situations where international competition is involved.

Lastly, subsidies do not always have equal benefits to both developed and developing nations. The developed nations usually have more money to fund extensive subsidies and hence have a comparative edge in the new green sector. Developing countries, on the contrary, might

not be able to offer such assistance, which would further deepen the economic inequality in the world.

Critical Evaluation

Green subsidies are only effective in internalising environmental externalities depending on a number of factors. To begin with, design and targeting are vital. The subsidies must be focused on technologies and industries that have the highest potential in terms of environmental impact and sustainability in the long-term. Inefficiency and unintended consequences may result due to poorly designed subsidies.

Second, subsidies suit best when used in conjunction with other policy tools. As an illustration, subsidies can be paired with carbon price or environmental policies to establish a more holistic policy framework that would not only promote clean options but also prevent harmful practices. In the absence of such integration, subsidies might not be adequate to yield considerable environmental gains.

Third, global coordination would be necessary to reduce distortions in trade, and to avoid subsidy wars. In order to make it aligning the policies among nations, cooperative approaches, such as common standards or multilateral agreements can be employed so as to minimise tension within the international trading mechanism.

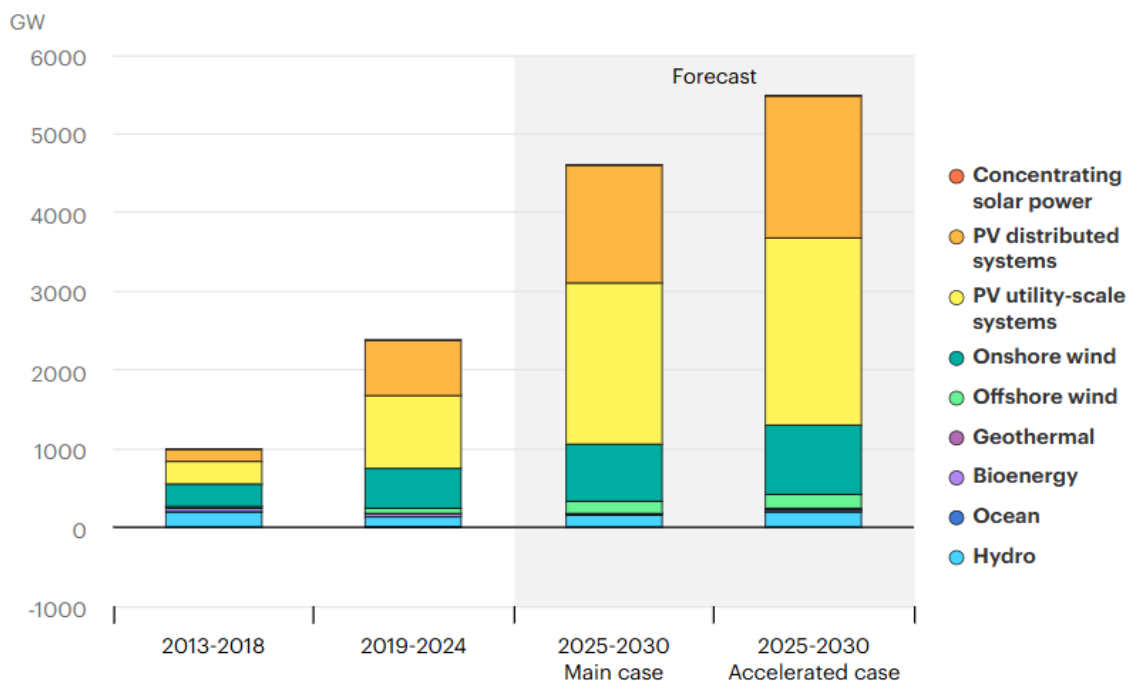
Fourth, it is important to take into account the issues of equity. The fact that the positive aspects of green subsidies are better distributed and that all the countries will be able to take part in the shift to the sustainable economy may be contributed by financial and technical assistance to the developing countries.

Conclusion

In conclusion, green subsidies are a powerful tool in environmental sustainability and internalisation of the good externalities of clean technologies. They are able to speed up the uptake of renewable energy, innovate and help develop the economy in the new industries. However, they are not effective enough because of some problems, such as distorting the market, spending fiscal resources, and the risk of international competition of subsidies.

Subsidies need to be properly designed, well-targeted, and incorporated into wider policy frameworks, with the aid of international collaboration and equal distribution of resources.

Capacity growth by generation technology, World



Source : [International Energy Agency](#)

(C) Standards and Regulations.

One of the most direct and commonly used policy tools in combating environmental externalities both domestically and internationally are environmental standards and regulations. In contrast to market-based instruments like tariffs and subsidies, standards provide certain rules or limits on the environmental impact of products and processes of production. They could be product standards (e.g. restrictions on harmful chemicals in goods), process standards (e.g. restrictions on emissions during production) and performance standards (e.g. requirement of energy efficiency).

Environmental standards have a two-fold role in the context of international trade. On the one hand, they are a means of internalisation of externalities in the environment, since they are means of ensuring that the minimum environmental requirements are met by the producers. Conversely, they affect trade flows in that access into the markets is stipulated by the environmental regulations of the country of importation since imported goods are obliged to meet the regulations of such countries.

One notable case is the Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH) Regulation enacted by the European Union. REACH obliges companies to recognize and address the risks posed by the chemicals that are employed in their products and processes. It puts the onus of proving safety of substances on the firms to prove that their substances are safe to the human health and the environment. This rule has far reaching consequences on the local manufacturers and the foreign exporters who want to enter the European market.

Role of Standards in Internalising Externalities

Theoretically, environmental standards are applicable to market failure through direct limitation of harmful practices. Instead of using price mechanisms, such as taxes, standards place quantitative or qualitative restrictions on pollution and environmental harm. To illustrate, emission standards are the maximum permissible levels of pollutants, and product standards can prohibit or limit the use of some dangerous materials.

Through such limits, standards make sure that firms internalise some of the environmental cost of their activities. To meet the regulations, firms would have to invest in cleaner technologies, use safer inputs or alter the production processes. This helps in minimizing negative externalities of production and consumption and brings the market nearer to a socially optimal solution.

Moreover, the standards can be used to deal with the issue of information asymmetry where consumers do not have access to information concerning the environmental impact of products. Examples of such schemes include eco-labelling and certification schemes, which give consumers information regarding sustainability allowing them to make knowledgeable decisions. This provides market incentives to firms to practice environmentally friendly practices, which strengthens the internalisation process even more.

Benefits of Environmental Standards.

One of the key advantages of environmental standards is their ability to directly control pollution. In contrast to market-based instruments, which use price signals to affect behaviour, standards provide definite and enforceable thresholds of environmental damage. This renders them especially useful in the case when environmental harm is to be carefully contained, e.g. in the case of poisonous materials or in the case of highly sensitive ecosystems.

The other advantage is that standards have the ability of enhancing the quality and safety of products. Laws like REACH provide that products that enter the market comply with high health and environmental standards. This ensures not only consumer and ecosystem protection, but also boosts product and market trust. An increase in standards has the potential to increase the quality of goods, which can advantage consumers in the long-term.

Standards may also help in the diffusion of cleaner technologies. Companies that are interested in penetrating into the markets that have stringent environmental laws will require to upgrade their production and practice more sustainable ways. This may eventually result in upgrading of technology and better environmental performance of various industries even in the exporting countries.

Moreover, standards bring about assurance and transparency. They establish certain rules that minimize ambiguity to both firms and regulators. This will aid in compliance and enforcement, especially where it is hard or impractical to measure environmental harm using pricing mechanisms.

Limitations and Challenges

Although environmental standards have their advantages, they have a number of critical constraints, especially in international trade. Among the most important issues is the fact that they may serve as trade barriers that are invisible. Although the justification of standards can be based on environmental or health considerations, they can be made or enforced in a manner that has an unequal impact on overseas manufacturers. It may limit access to the market and lessen competition, which is accused of protectionism.

This is especially a problem with developing countries. They might not be in a position to meet the high environmental standards because they might be deficient in technical capacity,

infrastructure and financial resources. Consequently, the exporters in these countries will either be charged more, or locked out of some markets. This calls into question fairness and equity in the international trading system.

The other key constraint is that the cost of compliance with environmental standards is high. Companies might be required to invest in new technologies, change the production processes, or undertake comprehensive tests and certification processes. These may be very large costs particularly to small and medium-sized enterprises (SMEs). There might be situations where the compliance cost is more than the benefit of entering specific markets.

Furthermore, standards can cause inefficiencies in relation to market-based instruments. Since they bring about uniform requirements, they do not give the firms the flexibility to decide on the most cost effective method of minimizing environmental harm. This may lead to an increase in the total expenditure in attaining a certain level of environmental protection.

The difficulty of harmonisation and coordination between countries exists as well. Various nations have various environmental standards, which can provide a complicated and fragmented regulating environment. This adds the strain on companies that have to conduct business in diverse markets since they are forced to meet different requirements. Absence of harmonisation can also diminish the efficacy of standards to solve international issues of environmental concern.

Further, the implementation of standards may be challenging, especially in countries where the institutions are weak. Lack of efficient monitoring and enforcement mechanisms can result in the standards existing on paper, restricting their ability to have any effect on the real environmental outcomes.

Critical Evaluation

The success of the environmental standards in internalising the externalities also lies much in the way the standards are designed, implemented and co-ordinated internationally. Properly designed standards based on scientific findings and implemented in a transparent manner can greatly minimize environmental degradation. Nevertheless, inadequately developed standards can impose unwarranted trade restrictions, and economic inefficiencies.

Standards are a more direct and immediate method of curbing environmental harm as compared to tariffs and subsidies. They are especially applicable to severe or irreversible environmental risks, and price-based instruments might not be adequate. They are, however, not very flexible and may affect trade limitations, which restrict their effectiveness.

Other policy instruments should be used to supplement standards so as to improve on their effectiveness. An example of this is a combination of standards and clean technology subsidies which can help in assisting firms to meet requirements of the regulations at a low cost. In the same breath, cross border cooperation can be achieved through agreements and standard congruencies and this will help in eliminating trade frictions and promote uniformity.

The developing countries must also be assisted. These countries can be assisted by offering them technical aid, building capacity, and financial resources to meet the environmental requirements and be more involved in international trade.

Conclusion

To sum up, regulations and standards of the environment are a potent means to tackle directly the environmental externalities, and guarantee the minimum degrees of environmental protection. They are now in a position to control the level of pollution, improve on the quality of their products and also encourage the use of cleaner technologies. However, their use in

international trade has serious problems, including the risk of hidden trade barriers, undue compliance costs, and unfair impacts across countries. Though standards are important in environmental governance, they need to be properly designed, implemented in a fair manner and coordinated internationally.

5. Comparative Case Studies

This section discusses three comparative case studies of the real-life effectiveness of the policy tools related to trade in internalising the environmental externalities which are carbon border policies in the European Union, renewable energy subsidies in large economies such as the United States and China and environmental standards introduced by the European Union. These examples demonstrate the practical effectiveness of tariffs, subsidies, and standards and emphasize their successes and shortcomings in various settings.

Case 1: EU Carbon Border Policy.

Carbon Border Adjustment Mechanism (CBAM) by the European Union is one of the most important efforts to incorporate the environmental goals into trade policy. The CBAM imposes a price on the imports of selected commodities such as steel, cement, aluminium, and fertilisers, which have a price depending on the embedded emissions. It is strongly associated with the domestic pricing system of carbon in the EU, which guarantees that imported goods are subjected to a comparable amount of environmental expenses as EU-produced goods.

Regarding its impact on imports, the CBAM will shift the trade patterns by making the carbon-intensive goods in the countries with weaker environmental regulations more expensive. The exporters to the EU are motivated to cut down on the emissions so as to stay in the market.

Early indications and projections indicate that companies that serve the EU market have started to adjust by enhancing efficiency in production, and embracing cleaner technologies. This illustrates the possibility of border measures to scale environmental standards past national borders.

In terms of emissions, the CBAM will help prevent carbon leakage and help cut down emissions in the world. The policy would discourage the offshoring of the polluting industries to other countries by making sure that the costs of carbon are passed on the price of imports, which would go to other countries and encourage a cleaner production elsewhere in the world. But its net effect on global emissions is not yet clear as it is restricted to a small number of sectors and relies on the actions of trading partners.

The CBAM has been criticized, despite its possible advantages. Countries developing also complain that it adds extra expenses to their exports and can inhibit economic growth. It is also questioned whether it is compatible with the rules of the World Trade Organization, especially on non-discrimination. Moreover, embedded emissions are difficult to measure, which is a practical challenge to implement. In general, the CBAM is a new method of internalising environmental externalities, but it should be designed, be transparent, and collaborate with other countries.

Case 2: United States and China Renewable Subsidies.

Subsidies have been important in furthering the development of renewable energy in the world, especially in some of the most important economies like the United States and China. The two nations have adopted massive subsidy schemes in an effort to promote the development of clean energy, albeit with different design and focus.

Policies in the United States, like the Inflation Reduction Act, have offered tax credits, grants, and monetary incentives on renewable energy generation, electric vehicles, and innovation of clean technology. These are meant to cut the amount of greenhouse gases and also enhance domestic production and energy security. This has seen the US record high increase in renewable energy capacity especially in solar and wind energy.

However, China has also taken a more state-oriented approach, as subsidies are coupled with industrial policy and long-term planning. The government subsidies have played a critical role in ensuring that China is the biggest producer of solar panels, wind turbines and batteries in the world. The Chinese subsidies have helped to bring a radical decrease in the cost of renewable technologies world wide and have helped benefit the domestic and international markets.

The increase in clean energy that has come as a result of these subsidy programs is impressive. There have been swift growths in renewable power generation capacity, reduction in the cost of clean technologies as well as increase in green industry investment in both countries. These results indicate that subsidies are effective in terms of extinguishing market failure in the arena of innovation and positive externalities.

Nevertheless, the application of subsidies has brought about problems as well. It is feared because it can lead to market distortion since the governmental assistance might favour some industries or firms above others. Moreover, the magnitude of subsidies has brought about concerns of subsidy race where nations engage in competition to win the subsidy.

This may result in inefficiencies and strains in international trade.

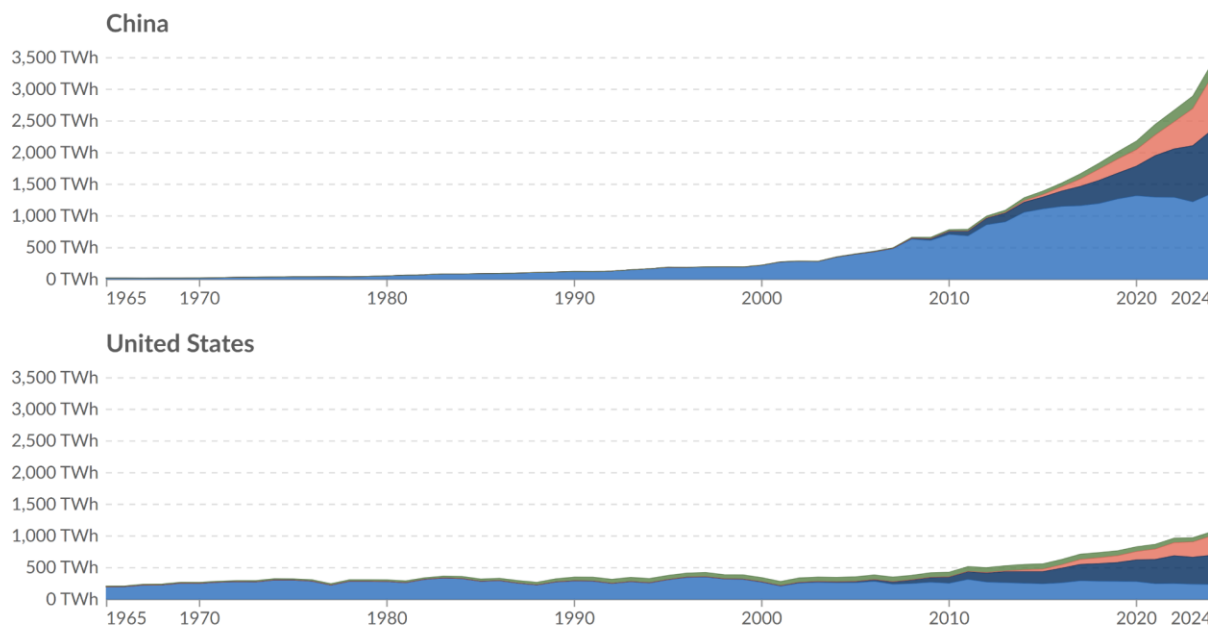
Additionally, subsidies in major economies can put smaller or emerging countries at a disadvantage since they do not have the financial capacity to offer subsidies. This casts doubt on equity and allocation of the benefits in the international shift to clean energy. In spite of them,

the examples of United States and China reveal the importance of subsidies to the development of environmental sustainability and technological progress.

Renewable electricity generation

Our World
in Data

Hydropower Wind Solar Other renewables



Data source: Energy Institute - Statistical Review of World Energy (2025)

OurWorldinData.org/renewable-energy | CC BY

Note: 'Other renewables' refers to renewable sources including geothermal, biomass, waste, wave and tidal. Traditional biomass is not included.

Sorce : Our World in Data

Case 3: Environmental Standards in the EU.

The European Union has been at the forefront in having strict environmental protection policies, such as the REACH, eco-design policies, and product safety policies among others. Such standards not only apply to national manufacturers but also to overseas exporters who are interested in entering the EU market and are, therefore, very influential in international trade.

The effect of this on the exports of the developing countries has been of great significance. On the one hand, EU standards have motivated exporters to enhance the quality of

their products and more sustainable and safer production methods. Companies which manage to meet these standards will be able to enter the high-value markets and increase their competitiveness. In this respect, EU regulations have the potential to be an agent of technological modernization and environmental enhancement in exporting nations.

Conversely, standards that are strict are usually expensive and technically challenging to adhere to, and may be overwhelming to companies in developing nations. Small and medium-sized businesses, especially, might find it difficult to comply with the complex regulation, to do the required testing and to get certifications. Consequently, part of the exporters will be locked out of the EU market and trade opportunities will be diminished and hence economic development may be compromised.

Also an issue is that the environmental standards can serve as non-tariff barriers to trade. Although they are acceptable on environmental and health grounds, they can have substantial effects on the flows of trade. There is a concern among critics that such standards might have a disproportionate influence on developing countries, which raises concerns about equity and fairness.

Irrespective of these difficulties, the EU environmental standards have brought about better environmental and safety practices worldwide. They are able to influence the production processes globally by creating high standards that can make companies move to more sustainable practices. This is known as the so-called Brussels effect and emphasizes the extent of EU regulations globally.

Comparative Insights

An analysis of these three cases shows that there is a lot to be learned about how effective trade-related policy tools are. Tariffs, as observed in the EU carbon border policy, are especially

beneficial in dealing with cross-border externalities and have some problems associated with trade rules and equity. As evidenced by the United States and China, subsidies can be very useful to encourage innovation and scaling up of clean technologies yet that they can cause distortions in the market and international competition. Examples of standards include EU regulations which offer direct control of the outcomes of the environment but pose a barrier to developing countries.

On the whole, these case studies give indications that no single policy tool can be used alone. All have their own strengths and limitations and their effectiveness varies depending on context, design and implementation. An integrated set of tariffs, subsidies and standards (with international co-operation) will perhaps be more effective in solving environmental externalities in a globalised economy.

6. Critical Evaluation

Tariff, subsidy, and standard analysis indicates that although all the instruments play a role in ensuring internalisation of environmental externalities, their actual performance is highly differentiated based on the circumstances, design, and implementation. This part presents a comparative assessment of these tools, and then a general discussion on main issues such as the variance between developed and developing nations, issues in enforcement, and limitations in the political economy.

Comparatively, the three instruments can be evaluated regarding their comparative effectiveness and limitations:

Instrument	Effectiveness	Limitations
Tariffs	Medium	Trade conflicts
Subsidies	High (innovation, transition)	Distortionary, expensive
Standards	High (direct environmental impact)	Trade barriers, inflexible

Tariffs, especially environmental tariffs, like carbon border adjustments, are moderately effective at internalising environmental externalities. Their main advantage is that they focus on cross-border problems, including carbon leakage, through harmonizing the costs of international imports on the environment with the domestic production standards. Nevertheless, coverage by sector, difficulties in measurement and reliance on international collaboration often restrict their effectiveness. In addition, the tariffs are very vulnerable to the disputes within the framework of the World Trade Organization because nations can dispute them as discriminatory or protectionist. This undermines their effectiveness as a long-term policy instrument.

Subsidies, in contrast, are very effective, especially when it comes to promoting innovation and easing the shift to cleaner technologies. Subsidies directly tackle the underinvestment that comes with positive externalities by reducing the cost of renewable energy and sustainable production methods. The experience of such major economies like the United States and China, has demonstrated that they have been able to scale up clean energy sectors and also reduce the cost of technology in the world. However, subsidies also have disadvantages of their own that should also be taken into consideration. They impose a tremendous burden on the

government and will lead to distortion in the market competition unless directed to it properly. In addition, the introduction of subsidy competition between countries can be the cause of the inefficiencies and tensions within the world trading system.

The most effective instrument of environmental externalities control, perhaps, is standards and regulations. They set firm limits on pollution and on the character of products in such a way that they can assure short-term compliance and the measurable environmental success. Indicatively, strict regulatory frameworks that have been enacted by the European Union have led to improvement of product safety and environmental performance. This effectiveness is however, expensive. Standards can also act as non-tariff barriers, restricting access by foreign producers to the market, and increasing compliance costs, particularly to developing country firms. They are also inflexible, which implies that they might not accomplish environmental objectives in the most cost-effective way.

Developed vs Developing Countries.

Amongst the essential considerations to be made in the assessment of these instruments is the fact that they have an unequal impact on the developed and developing countries. The developed countries tend to be much organized regarding institutions, financial resources and technological facilities are at an advanced stage. This would allow them to design, implement and enact complicated policy tools in a more effective way.

Developing countries on the other hand tend to be constrained by factors such as administrative weaknesses, inefficient infrastructure and other developmental priorities. They therefore may not be able to comply with strict environmental demands or they might not adapt to environmental tariffs practiced by developed countries. To illustrate this, carbon tariffs and their products can be locked out of the high-value markets with high standards, and

competitiveness of the exports of developing economies can be reduced. The subsidies also help in highlighting this disparity.

The developed countries can sustain the massive financial aid that is offered to green industries but the developing countries may not be in a position to sustain the same due to lack of fiscal space. This puts a leveled playing field with richer countries benefiting in the new markets like clean technology and renewable energy.

These differences bring up significant issues of equality and justice in international environmental regulation. Devoid of proper support mechanisms (including, but not limited to, transfer of technology, financial aid, and capacity building) the environmental policies that are linked to trade may only contribute to the existing inequalities.

Enforcement Challenges

Enforcement is a critical aspect of the effective operation of all three instruments, but it is also a major problem in practice. In the case of tariffs, the implementation should entail proper quantification of the environmental effects like the carbon emission contained in the goods that are traded. This is technically complicated and can be limited by data and methodological differences.

Although easier to apply within the country, subsidies have to be monitored to make sure they do not result in abuse and ineffectiveness. There is also a need to determine whether or not subsidised industries will continue to be reliant or can become self-reliant in the long run.

Standards although direct and clear have a problem of enforcement issues to do with monitoring and compliance. Standards can be in law and not in practice in countries that have weak regulatory institutions. In developed economies, compliance may be challenging even in the complex world-wide supply chains.

On the international level, the lack of a central authority further complicates enforcement. Although organisations like the world trade organisation offer dispute resolution mechanisms, they are more concerned with trade regulations as opposed to environmental performance. This leaves loopholes in implementation of trade environmental policies.

Political Economy Issues

Political economy factors also influence the design and implementation of environmental policies related to trade. Governments have to strike a balance between environmental goals and economic stakes, including competitiveness, employment, and growth. This tends to result in tradeoffs that diminish the efficiency of policy instruments.

Indicatively, the domestic industry lobbying can affect the tariffs, which leads to selective application or protectionism. Subsidies are also an easy target by politicians, who can choose who gets financial assistance according to their strategic or electoral interests instead of its environmental efficacy. This may result in poor utilization of the state resources and less influence of policy.

Although standards tend to be based on scientific evidence, they may be influenced by political and economic interests. It is possible that countries can use more stringent standards to safeguard their domestic industries or to react to social demands, when less restrictive ones could have the same environmental effects.

The other major problem is the uncoordinated approach internationally. The externalities related to the environment, i.e., climate change, are international in character and must be addressed on a collective level. Nevertheless, coordination is hard because of the differences in the priorities of the countries, their levels of development, and political regimes. Consequently,

the policies tend to be adopted unilaterally, diminishing their overall effectiveness and exposing them to the possibility of conflict.

Overall Assessment

Comparing the three tools, one can easily know that all of them have their own role to play in the field of environmental externalities. Tariffs are useful in regulating the cross-border impacts but are disadvantageous because of the trade wars as well as administrative challenges. Innovation and acceleration of uptake of sustainable technologies can be very successful using subsidies, but are linked to fiscal and competition challenges. Standards directly and quantifiably beneficial to the environment but may lead to barriers to trade and high compliance costs.

Environmental externalities cannot be completely internalised by one instrument. Instead, an additional policy of integrating tariffs, subsidies and standards will be more fruitful. International collaboration, powerful institutional structures, and ways of dealing with equity issues should underpin such an approach.

Conclusion of Critical Evaluation

In conclusion, despite the fact that policy tools applicable to trade can provide useful means of correcting the externalities of the environment, their application is necessarily limited by the economic, institutional and political factors. An integrated-balanced system of policy, sensitive to the demands of the developed and the developing countries is essential towards achieving sustainable and equal outcomes in the global economy.

7. Policy Implications

In the discussion of the tariffs, subsidies and standards, it is highlighted that whereas each of the instruments can internalise the environmental externalities, in general each of them is

limited in its effectiveness by the design constraints, disproportionate impacts, and coordination problems. Therefore, the balanced and broad policy approach is required to make them more efficient and ensure that the objectives of the environment and trade go hand-in-hand. The section presents some important policy recommendations, which are enhanced international coordination, strategic integration of tools and developing country support.

Improvement of Co-ordination in World Trade Organization.

One of the main policy implications is that there is a necessity of closer integration between trade and environmental governance systems, especially within the World Trade Organization (WTO). Today, WTO regulations are more inclined towards the encouragement of free and fair trade with the environmental issues being addressed by the means of the few exceptions. This generates strains when nations strive to enact trade-related environmental policies like carbon tariffs or environmental policies.

In order to deal with this problem, the WTO provisions regarding environmental protection should be clarified and extended. This may include working out more explicit rules on the application of environmental tariffs and subsidies so that they are not abused as protectionist instruments. It would also cut down on controversies and bring more openness by the setting of internationally accepted methodologies of measuring environmental impacts like the carbon content of goods.

Moreover, WTO could be more active in terms of promoting dialog and collaboration among member states. The organisation can also facilitate harmony in policies and mitigate the chances of trade wars by encouraging multilateral agreements on environmental trade measures. Better congruence of trade policies with international environmental policies would enhance the effectiveness of policy interventions.

Integrating Policy Instruments to be more effective.

The other important implication is that no one instrument alone will be adequate to tackle environmental externalities. Rather, policymakers need to embrace a complementary policy that integrates tariffs, subsidies and standards so that they can use their advantages.

An illustration is the tariffs like carbon border adjustments can be implemented to deal with cross-border externalities and stop carbon leakage. Simultaneously, the subsidies may facilitate the emergence and implementation of clean technologies that will lower compliance expenses of companies. Standards and regulations can ensure minimum levels of environmental protection by directly limiting harmful activities.

These instruments can be used in conjunction with each other to form a more effective policy framework. Varying Policy Instruments to be more effective. As an example, subsidies may be used to assist companies to switch to cleaner production modes, standards ensure that firms comply with the environmental standards and tariffs keep the firms competitive through the equalisation of costs between local and foreign firms. This combined strategy could be used to improve efficiency and effectiveness.

Nevertheless, it is necessary to be careful when designing and sequencing the policies. The policymakers should make sure that these instruments are coordinated and do not conflict. To give an example, environmentally damaging activities should not be subsidised and standards should be made in a way that would reduce trade barriers that are unnecessary. Honesty and uniformity are also essential in ensuring trust between trading partners.

Supporting Developing Countries

One such policy issue of high priority is the need to deal with the inequitable effects of trade and environmental policies on the developing nations. As noted earlier in this paper,

emerging economies are generally disadvantaged in meeting the environmental tariffs and standards, and in enforcing subsidy schemes because of inadequate financial resources and institutions.

Developed countries and international organisations ought to offer financial and technical support to developing countries in order to make them fair and inclusive. This can be in the form of subsidized adoption of clean technology, subsidized development of infrastructure and capacity building projects to enhance regulatory and enforcement mechanisms.

Of great significance is the transfer of technology that helps the developing countries to embrace sustainable production processes. Countries can cut down emissions without having to undermine economic growth by enabling access to advanced technologies. Furthermore, the individual responsibilities and differentiation of implementation schedules may be used to accommodate different stages of development.

Arrangements of international cooperation arrangements such as the arrangements of the Paris Agreement could be very helpful in ensuring equitable burden sharing. Such agreements can be used to ensure that global sustainability objectives are achieved in a balanced and coordinated way through aligning of trade related environmental policies.

Increasing Transparency and Accountability.

The second important implication is that there is need to be more transparent and accountable in how they use trade-related environmental instruments. In order to avoid misunderstandings and to build up trust with trading partners, governments should communicate effectively the goals, design, and expected results of policies.

The mechanisms of monitoring and evaluation need to be reinforced to determine the effectiveness of policy instruments with time. This would be tracking of environmental

performance, economic and distributional performance. Policy changes can be informed by such assessments and make sure that actions are sustainable.

Conclusion of Policy Implications

Lastly, to enhance the efficacy of the policy instruments related to trade, a need to have a coordinated and integrated approach that is inclusive. The introduction of the role of the World Trade Organization, tariff integration, subsidies and standards and specific target assistance to the developing nations are essential steps towards sustainable and fair outcomes. With the coordination of the policies of trade and environment through a coherent framework, governments will be in a position to improve the externalities of the environment without sacrificing the gains of international trade.

8. Conclusion

This paper has explored the usefulness of the trade relevant policy tools such as tariffs, subsidies and standards in internalising environmental externalities in a globalised economy. As shown in the analysis, these instruments, though they offer valuable mechanisms of addressing market failure, have low internalisation of environmental costs and are highly context-specific.

In the most basic sense, environmental externalities are a typical instance of a market failure, where market prices do not capture the social cost of production and consumption. This issue is exacerbated by trade through the growth of production, an increase in the utilization of resources, and the movement of activities that produce pollution across the border. Consequently, policy interventions that can reconcile the social welfare and the private incentives are evidently needed. One such avenue is the trade-related instruments which are seen to integrate environmental issues into international economic interactions.

In the assessment of individual instruments, their contributions and limitations are identified. Tariffs, especially environmental tariffs like carbon border adjustments are useful in dealing with cross-border externalities and carbon leakage. They spread the scope of domestic environmental policies and promote cleaner production practices internationally. Nonetheless, this is limited in their effectiveness due to the measurement, compliance, and possible conflicts issues within the World Trade Organization framework. Also, their influence on the developing countries poses the issue of equity and fairness.

The subsidies, particularly those that promote renewable energy and clean technologies have been very effective in creating innovations and speeding up the process of transition to a low-carbon economy. The fact that major economies like the United States and China have been involved in the scaling up of clean industries and cost reduction due to technological advancement is evidenced. However, subsidies have high fiscal costs, market distortion risks and subsidy races as a means of international competition. These aspects restrict their effectiveness and cast doubt on the sustainability in the long term.

Standards and regulations provide the simplest method of regulating environmental externalities through the establishment of explicit limits of pollution and product features. The European Union policies provide an example of how high standards may enhance environmental performance and product quality at the global level. Nonetheless, they are very rigid, expensive to comply with, and may serve as non-tariff protection, which poses a major challenge especially to developing countries in their quest to access the international markets.

Among the key discoveries of this study is the fact that the practical usefulness of these instruments is not simply dictated by their theoretical construction but also the practical side, i.e., the possibility to enforce and the quality of the institution and the political will. Policies that are

well formulated can be negated by weak institutions whereas strong institutions can enhance their effectiveness. Similarly, lack of reliable data and surveillance systems can limit the utility of tariffs and norms and, a lack of enforceable systems, can reduce the utility of subsidy programs.

Another critical factor is the need for global cooperation. Transnational in nature are the environmental externalities, in particular, climate change. Though it has provided a system through which trade relations can be managed, and agreements such as the Paris Agreement are also aimed at coordinating the environment action, integration of these frameworks is still incomplete. There should be a stronger coordination between trade and environmental governance to ensure the policy instruments are useful as much as possible and that conflicts are reduced to the minimum.

It also calls on the analysis the need to address distributional and equity concerns, including the difference in the capacity of developed and developing countries. Devoid of proper support, the environmental policies relating to trade might heavily hurt the developing economies, as they will not be able to engage in international trade and to attain sustainable development. This brings in the importance of having complementary efforts like financial support, transfer of technology and capacity-building efforts.

As shown by the results of the research, trade-based policy instruments are effective tools of environmental protection, but they also have inherent limitations in the scope of their work. Three crucial factors that entail system design and enforcement systems and international measures taken in regulating its operations are what make the system successful. Environmental costs cannot be internalised solely by use of a single instrument, and it is not likely that any one of a variety of instruments alone could produce the best results.

Global environmental governance in its entirety should be implemented using trade related policy instruments since they are indispensable requirements of this process. To deal with the intricate and interrelated issues of trade and environmental sustainability, a holistic approach that incorporates tariffs, subsidies, and standards into an integrated international system is necessary. The international economy requires policy makers to formulate regulatory structures that combine economic motivation with international collaboration and cooperation in ensuring the sustainable economic growth along with environmental conservation.

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